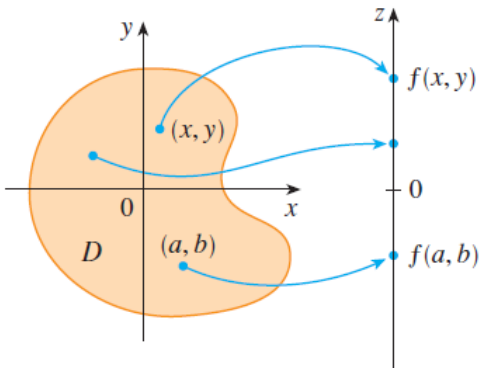
In these notes we review a few results from multivariable calculus that we will need in our study of differential equations.

1 Functions of two variables

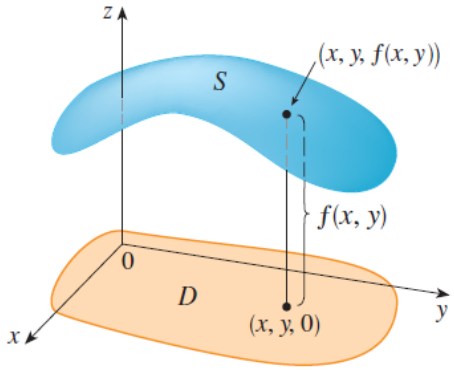
Consider a function $f : D \subseteq \mathbb{R}^2 \to \mathbb{R}$, which is called a *real-valued function of two variables*. Such a function takes each point $(x,y) \in D$ to a real number $z = f(x,y)$. Its graph

$$\{(x,y,z) : z = f(x,y)\} \subseteq \mathbb{R}^3$$

is a surface S in three-dimensional space and can be visualized as lying directly above or below its domain D in the xy-plane.



(a) A mapping diagram for a function $f : D \subseteq \mathbb{R}^2 \to \mathbb{R}$.



(b) The graph of a function $f : D \subseteq \mathbb{R}^2 \to \mathbb{R}$.

Figure 1: Two ways to visualize a real-valued function of two variables.

Example 1.1. Consider the function $f(x,y) = \sqrt{9-x^2-y^2}$. The domain is the set of all (x,y) such that $9-x^2-y^2 \geq 0$ or, equivalently, $x^2+y^2 \leq 9$, which is the interior of the unit disc of radius 3 centered at the origin in the xy-plane. Note that any function $f(x,y)$ which depends on x and y only in the combination x^2+y^2 will have a graph that is rotationally-symmetric about the z-axis. [1]

[1] To see why this is true, suppose we rotate any point $(x,y) \in D$ by an angle θ about the z-axis and denote the resulting point (x',y'). Then

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} = \begin{bmatrix} x\cos(\theta) - y\sin(\theta) \\ x\sin(\theta) + y\cos(\theta) \end{bmatrix},$$

so we have

$$\begin{aligned} (x')^2 + (y')^2 &= (x\cos(\theta) - y\sin(\theta))^2 + (x\sin(\theta) + y\cos(\theta))^2 \\ &= x^2\cos^2(\theta) + y^2\sin^2(\theta) - 2xy\sin(\theta)\cos(\theta) \\ &\quad + x^2\sin^2(\theta) + y^2\cos^2(\theta) + 2xy\sin(\theta)\cos(\theta) \\ &= x^2(\sin^2(\theta) + \cos^2(\theta)) + y^2(\sin^2(\theta) + \cos^2(\theta)) \\ &= x^2 + y^2, \end{aligned}$$

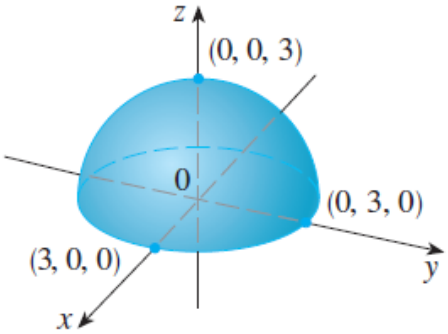
and therefore $f(x',y') = f(x,y)$.

To use this fact to help us sketch the graph, we can consider the intersection of the graph of f with the xz-plane by setting $y = 0$:

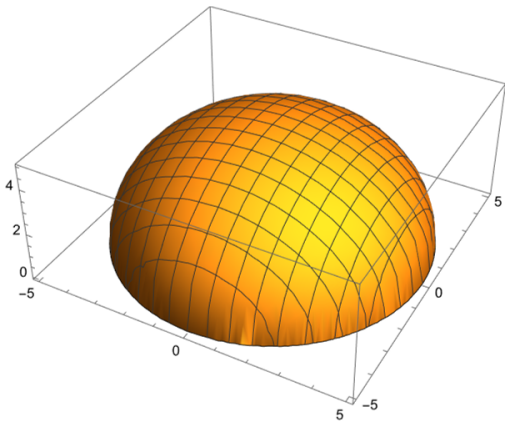
$$f(x,0) = \sqrt{9-x^2}, \quad -3 \le x \le 3,$$

which is a semi-circle of radius 3 centered at the origin in the upper half xz-plane. Rotating this about the z-axis then gives the graph of f, which is the upper hemisphere of a sphere of radius 3 centered at the origin (see Figure 2a below). The graph of this function can be generated in *Mathematica* as follows (see Figure 2b below):

```
In[1]:=Plot3D[(25-x^2-y^2)^{1/2},{x-5,5},{y,-5-5}]
```



(a) Graph of the function $f(x,y) = \sqrt{9-x^2-y^2}$.



(b) Graph of the function $f(x,y) = \sqrt{9-x^2-y^2}$ generated by *Mathematica*.

Figure 2: Graph of the function $f(x,y) = \sqrt{9-x^2-y^2}$.

Exercise 1.2. Sketch the graph of each of the following functions. Check your work by comparing with the graphs generated by *Mathematica*.

(a) $f(x,y) = 25 - x^2 - y^2$

(b) $g(x,y) = 25 - \sqrt{x^2 + y^2}$

2 Limits and Continuity

Recall that the distance between two points (x_1, y_1) and (x_2, y_2) in $\mathbb{R}^2$ is given by $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$.

Definition 2.1. Let $R > 0$. The *punctured open disc in* $\mathbb{R}^2$ *of radius* R *centered at* (x_0, y_0) is the set

$$\{(x,y) : 0 < \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} < R\} \subseteq \mathbb{R}^2.$$

This is a disc of radius R whose center is (x_0, y_0), with the point (x_0, y_0) itself removed.

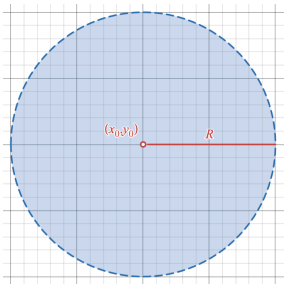
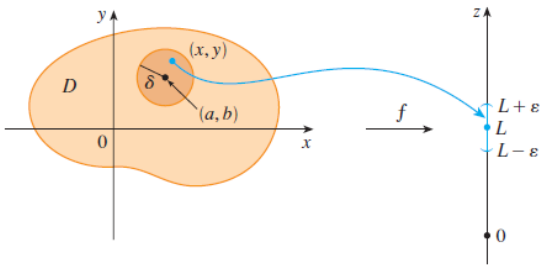


Figure 3: A punctured open disc of radius R at (x_0, y_0).

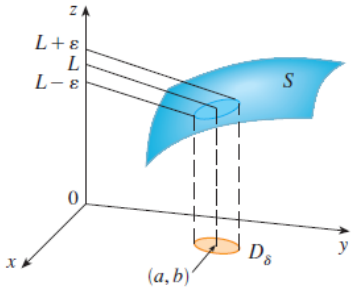
Given a function $f : D \subseteq \mathbb{R}^2 \to \mathbb{R}$ defined on a punctured disc D of some positive radius centered at $(x_0, y_0) \in \mathbb{R}^2$ and some real number L, we say that L *is the limit of* $f(x,y)$ *as* (x,y) *approaches* (x_0, y_0) and write

$$\lim_{(x,y)\to(x_0,y_0)} f(x,y) = L$$

if we can make $f(x,y)$ arbitrarily close to L by taking (x,y) sufficiently close to (x_0, y_0); that is, if for every $\varepsilon > 0$ there exists $\delta > 0$ such that $(x,y) \in D$ and $0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta$ implies $|f(x,y) - L| < \varepsilon$.



(a) Illustration of $\lim_{(x,y)\to(a,b)} f(x,y) = L$ in terms of a mapping diagram.



(b) Illustration of $\lim_{(x,y)\to(a,b)} f(x,y) = L$ in terms of the graph of f.

Figure 4: Different ways of visualizing $\lim_{(x,y)\to(a,b)} f(x,y) = L$.

Definition 2.2. We say that f *is continuous at* (x_0, y_0) if f is defined at (x_0, y_0) and $\lim_{(x,y)\to(x_0,y_0)} f(x,y) = f(x_0, y_0)$ and we say that f is *continuous* if it is continuous at each point in its domain.

Example 2.3. Suppose $f(x,y) = g(x)$, where $g(x)$ is a function of one variable such that $\lim_{x\to x_0} g(x) = L$. It follows that, for any y_0 such that (x,y_0) is in the domain of f, we have

$$\lim_{(x,y)\to(x_0,y_0)} f(x,y) = \lim_{x\to x_0} g(x) = L.$$

That is, the limit of f as a function of two variables is the same as its limit as a function of one variable. To prove this, let $\varepsilon > 0$. Since $\lim_{x\to x_0} g(x) = L$, there exists a corresponding $\delta > 0$ such that $0 < |x-x_0| < \delta$ implies $|g(x)-L| < \varepsilon$. Since

$$0 < |x-x_0| = \sqrt{(x-x_0)^2} \le \sqrt{(x-x_0)^2+(y-y_0)^2},$$

if we take

$$\sqrt{(x-x_0)^2+(y-y_0)^2} < \delta,$$

then we will have $0 < |x-x_0| \le \sqrt{(x-x_0)^2+(y-y_0)^2} < \delta$ [2] and therefore $|g(x)-L| < \varepsilon$. But $g(x) = f(x,y)$, so this implies $|f(x,y)-L| < \varepsilon$. It follows from this example that if $g(x)$ is continuous, then so is $f(x,y)$. Similar considerations show that f is continuous if $f(x,y) = h(y)$.

Limit Laws

In one-variable calculus, calculations of limits of many complicated functions can be reduced to calculations of limits of simpler functions by the limit laws. The proofs of the limit laws in one-variable calculus all follow from the following three properties of the one-dimensional distance function $d(x_1,x_2) = |x_1-x_2|$:

(i) (postive-definiteness) $|x_1-x_2| \ge 0$ and $= 0$ if and only if $x_1 = x_2$.

(ii) (symmetry) $|x_1-x_2| = |x_2-x_1|$ for all x_1, x_2.

(iii) (triangle inequality) $|x_1-x_2| \le |x_1-x| + |x-x_2|$ for all x_1, x_2.

Since the two-dimensional distance function $\sqrt{(x_1-x_2)^2+(y_1-y_2)^2}$ also satisfies these three properties, the same proofs of the limit laws in one-variable calculus show that the same limit laws all hold for real-valued functions of two variables. The proof of the Squeeze Theorem is also the same as in one-variable calculus. It follows that sums, differences, products, quotients (when the denominator is nonzero), and compositions of continuous functions are all continuous. For example

$$\sum_{i,j=1}^{n} c_{ij}x^i y^j, \quad \sin(xy), \quad \frac{e^{x^2-y^2}}{x^2+y^2+1}, \quad \text{etc.}$$

are all continuous on $\mathbb{R}^2$.

Exercise 2.4. Where is the function $f(x,y) = \ln(x^2+y^2-4)$ continuous?

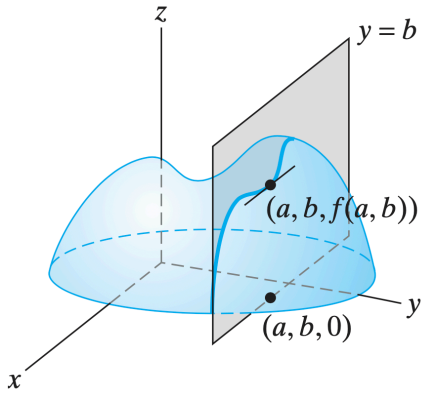
[2] This just says that if (x,y) is within an open disc of radius δ centered at (x_0,y_0), then x must be in an open interval centered at x_0 of radius δ.

3 Partial Derivatives

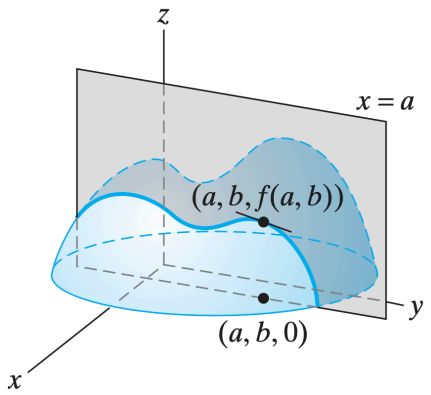
Given a real-valued function $f(x,y)$ of two variables, we can hold one variable constant and calculate the instantaneous rate of change of f with respect to the other variable. This is called a *partial derivative*. The partial derivatives with respect to x and y are denoted, respectively, by

$$\frac{\partial f}{\partial x} = \lim_{h\to 0} \frac{f(x+h,y)-f(x,y)}{h},$$
$$\frac{\partial f}{\partial y} = \lim_{h\to 0} \frac{f(x,y+h)-f(x,y)}{h}.$$

Another common notation is $f_x := \frac{\partial f}{\partial x}$ and $f_y := \frac{\partial f}{\partial y}$. Geometrically, $\frac{\partial f}{\partial x}(a,b)$ is the slope of the tangent line to the curve obtained by intersecting the surface $z = f(x,y)$ with the plane $y = b$ at the point $(a,b,f(a,b))$. Similarly, $\frac{\partial f}{\partial y}(a,b)$ is the slope of the tangent line to the curve obtained by intersecting the surface $z = f(x,y)$ with the plane $x = a$ at the point $(a,b,f(a,b))$.



(a) Geometric interpretation of $\frac{\partial f}{\partial x}(a,b)$.



(b) Geometric interpretation of $\frac{\partial f}{\partial y}(a,b)$.

Figure 5: Geometric interpretation of partial derivatives at a point.

To calculate $\frac{\partial f}{\partial x}$, say, one simply treats y as a constant and applies the usual derivative rules from one-variable calculus. One similarly calculates $\frac{\partial f}{\partial y}$ by treating x as a constant.

Example 3.1. If $f(x,y) = x^3 + x^2y - 2y^2$, then

$$\frac{\partial f}{\partial x} = 3x^2 + 2xy^3$$
$$\frac{\partial f}{\partial y} = 3x^2y^2 - 4y.$$

Exercise 3.2. Calculate the partial derivatives of the function $f(x,y) = e^x \sin(xy)$

Second partial derivatives

By allowing the point (x,y) at which we calculate the partial derivatives to vary, $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$ themselves becomes real-valued functions of two variables and we can similarly calculate their partial derivatives. This leads to four second partial derivatives:

$$\frac{\partial^2 f}{\partial x^2} := \frac{\partial f}{\partial x}\left(\frac{\partial f}{\partial x}\right), \quad \frac{\partial^2 f}{\partial x \partial y} := \frac{\partial f}{\partial y}\left(\frac{\partial f}{\partial x}\right), \quad \frac{\partial^2 f}{\partial y \partial x} := \frac{\partial f}{\partial x}\left(\frac{\partial f}{\partial y}\right), \quad \frac{\partial^2 f}{\partial y^2} := \frac{\partial f}{\partial y}\left(\frac{\partial f}{\partial y}\right)$$

These are also frequently denoted as

$$f_{xx} := \frac{\partial^2 f}{\partial x^2}, \quad f_{xy} := \frac{\partial^2 f}{\partial y \partial x}, \quad f_{yx} := \frac{\partial^2 f}{\partial x \partial y}, \quad f_{yy} := \frac{\partial^2 f}{\partial y^2}.$$

Example 3.3. The second partial derivatives of the function $f(x,y) = x^3 + x^2 y - 2y^2$ from Example 3.1 are

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}(3x^2 + 2xy^3) = 6x + 2y^3, \qquad \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial y}(3x^2 + 2xy^3) = 6xy^2,$$
$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial x}(3x^2y^2 - 4y) = 6xy^2, \qquad \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}(3x^2y^2 - 4y) = 6x^2y - 4.$$

Exercise 3.4. Compute the second partial derivatives of the function $f(x,y) = e^x \sin(xy)$ from Exercise 3.2

Equality of Mixed Second Partial Derivatives

For each of the previous two functions, the mixed second partial derivatives were equal: $f_{xy} = f_{yx}$. The next theorem shows that this was no coincidence.

Theorem 3.5 (Clairaut's Theorem)**.** *Suppose f is defined on a disk D that contains the point (a,b). If the functions $\frac{\partial^2 f}{\partial x \partial y}$ and $\frac{\partial^2 f}{\partial y \partial x}$ are both continuous on D, then $\frac{\partial^2 f}{\partial x \partial y}(a,b) = \frac{\partial^2 f}{\partial y \partial x}(a,b)$.*

Proof. For sufficiently small, non-zero values of h, consider the difference

$$\Delta(h) := [f(a+h,b+h) - f(a+h,b)] - [f(a,b+h) - f(a,b)].$$

If we let $g(x) := f(x,b+h) - f(x,b)$, then

$$\Delta(h) = g(a+h) - g(a).$$

By the Mean Value Theorem, there exists a number c between a and $a+h$ such that

$$\begin{aligned} g(a+h) - g(a) &= g'(c)h \\ &= h\left(\frac{\partial f}{\partial x}(c,b+h) - \frac{\partial f}{\partial x}(c,b)\right). \end{aligned}$$

Applying the Mean Value Theorem again, this time to $\frac{\partial f}{\partial x}$, we get a number d between b and $b+h$ such that

$$\begin{aligned} \frac{\partial f}{\partial x}(c,b+h) - \frac{\partial f}{\partial x}(c,b) &= \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)(c,d)h \\ &= \frac{\partial^2 f}{\partial y \partial x}(c,d)h. \end{aligned}$$

Combining these equations, we obtain

$$\Delta(h) = h^2 \frac{\partial^2 f}{\partial y \partial x}(c,d).$$

As $h \to 0$, then $(c,d) \to (a,b)$, so by continuity of $\frac{\partial^2 f}{\partial y \partial x}$ at (a,b) we have

$$\lim_{h \to 0} \frac{\Delta(h)}{h^2} = \lim_{(c,d) \to (a,b)} \frac{\partial^2 f}{\partial y \partial x}(c,d) = \frac{\partial^2 f}{\partial y \partial x}(a,b).$$

Similarly, by writing instead

$$\Delta(h) := [f(a+h,b+h) - f(a,b+h)] - [f(a+h,b) - f(a,b)]$$

and using the Mean Value Theorem twice and the continuity of $\frac{\partial^2 f}{\partial x \partial y}$ at (a,b), we obtain

$$\lim_{h \to 0} \frac{\Delta(h)}{h^2} = \frac{\partial^2 f}{\partial x \partial y}(a,b).$$

By uniqueness of limits, we conclude that

$$\frac{\partial^2 f}{\partial x \partial y}(a,b) = \frac{\partial^2 f}{\partial y \partial x}(a,b).$$

□

4 Differentiability

Recall that, for a real-valued function f of one variable, differentiability at a implies continuity at a, since

$$\lim_{\Delta x\to 0}[f(a+\Delta x)-f(a)] = \lim_{\Delta x\to 0}\left[\frac{f(a+\Delta x)-f(a)}{\Delta x}\Delta x\right] = \lim_{\Delta x\to 0}\frac{f(a+\Delta x)-f(a)}{\Delta x}\lim_{\Delta x\to 0}\Delta x = f'(a)\cdot 0 = 0.$$

However, if f is a real-valued function of two variables, the existence of partial derivatives at (a,b) is not a strong enough condition to guarantee continuity, as the next example shows.

Example 4.1. For example, consider

$$f(x,y) = \begin{cases} 1, & \text{if } xy = 0, \\ 0, & \text{otherwise.} \end{cases}$$

That is, this function takes the value 1 on the x and y axes, and 0 otherwise. It follows that

$$\frac{\partial f}{\partial x}(0,0) = \lim_{\Delta x\to 0}\frac{f(0+\Delta x,0)-f(0,0)}{\Delta x} = \lim_{\Delta x\to 0}\frac{0-0}{\Delta x} = 0,$$
$$\frac{\partial f}{\partial y}(0,0) = \lim_{\Delta y\to 0}\frac{f(0,0+\Delta y)-f(0,0)}{\Delta y} = \lim_{\Delta y\to 0}\frac{0-0}{\Delta y} = 0.$$

However, this function is not continuous at the origin, since the limit if we approach the origin along the x- or y-axis is 1, while if we approach the origin along the line $y = x$, the limit is 0. Hence, $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist.

For a real-valued function of one variable $y = f(x)$, differentiability at a is the same as the existence of a "best linear approximation" (e.g., the tangent line) to the graph of f at $(a, f(a))$. The latter is the key idea of differentiability of a function $z = f(x,y)$ of two variables at (a,b). For the function in the previous example, there is no "best linear approximation" (in this case, tangent plane) to the graph at the origin. No matter which plane we take through $z = 1$, it is not arbitrarily close to all nearby points on the graph of f.

Let's state precisely what is meant by "best linear approximation." We begin with a real-valued function of one variable, $y = f(x)$. If x changes from a to $a+\Delta x$, then the corresponding change in y is

$$\Delta y = f(a+\Delta x) - f(a).$$

If f is differentiable at a, then using the limit laws we can rewrite the definition of $f'(a)$ as follows:

$$\begin{aligned}
\lim_{\Delta x\to 0} \frac{\Delta y}{\Delta x} &= f'(a) \\
\lim_{\Delta x\to 0} \frac{\Delta y}{\Delta x} - f'(a) &= 0 \\
\lim_{\Delta x\to 0} \left[\frac{\Delta y}{\Delta x} - f'(a)\right] &= 0 \\
\lim_{\Delta x\to 0} \left[\frac{\Delta y}{\Delta x} - \frac{f'(a)\Delta x}{\Delta x}\right] &= 0 \\
\lim_{\Delta x\to 0} \frac{\Delta y - f'(a)\Delta x}{\Delta x} &= 0
\end{aligned}$$

If we define

$$\varepsilon := \frac{\Delta y - f'(a)\Delta x}{\Delta x},$$

which we think of as a function of Δx, then the manipulations above show that differentiability of f at a is equivalent to the statement that

$$\Delta y = f'(a)\Delta x + \varepsilon\Delta x, \quad \text{where } \lim_{\Delta x\to 0} \varepsilon = 0. \tag{4.2}$$

This is the precise statement that $f'(a)\Delta x$ is the "best linear approximation" to Δy.

Now consider a function of two variables, $z = f(x,y)$, and suppose x changes from a to $a+\Delta x$ and y changes from b to $b+\Delta y$. The corresponding change in z is then

$$\Delta z = f(a+\Delta x, b+\Delta y) - f(a,b).$$

In analogy with (4.2), we define differentiability of a function of two variables as follows:

Definition 4.3 (Differentiability of a function of two variables)**.** If $z = f(x,y)$, then f is *differentiable* at (a,b) if Δz can be expressed in the form

$$\Delta z = \frac{\partial f}{\partial x}(a,b)\Delta x + \frac{\partial f}{\partial y}(a,b)\Delta y + \varepsilon_1\Delta x + \varepsilon_2\Delta y, \tag{4.4}$$

where $\lim_{(\Delta x,\Delta y)\to(0,0)} \varepsilon_1 = \lim_{(\Delta x,\Delta y)\to(0,0)} \varepsilon_2 = 0$.

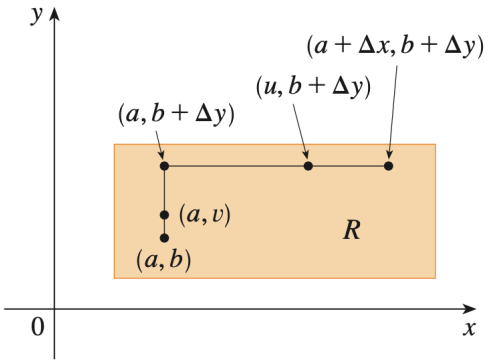
Note that $z = f(a,b) + \frac{\partial f}{\partial x}(a,b)\Delta x + \frac{\partial f}{\partial y}(a,b)\Delta y$ is the equation of the tangent plane to the graph of $z = f(x,y)$ at $(a,b,f(a,b))$, so this expresses the condition that the tangent plane is the best linear approximation to Δz. Note also that it is immediate from (4.4) that $\lim_{(\Delta x,\Delta y)\to(0,0)} \Delta z = 0$, so that differentiability implies continuity, as is the case for one-variable functions.

The next theorem gives a sufficient condition for differentiability.

Theorem 4.5. *If the partial derivatives* $\frac{\partial f}{\partial x}$ *and* $\frac{\partial f}{\partial y}$ *exist on an open set containing* (a,b) *and are continuous at* (a,b)*, then* f *is differentiable at* (a,b)*.*

Proof. Referring to the figure below, we write

$$\begin{aligned}\Delta z &= f(a+\Delta x, b+\Delta y) - f(a,b)\\ &= f(a+\Delta x, b+\Delta y) - f(a,b+\Delta y) + f(a,b+\Delta y) - f(a,b)\\ &= [f(a+\Delta x, b+\Delta y) - f(a,b+\Delta y)] + [f(a,b+\Delta y) - f(a,b)].\end{aligned} \tag{4.6}$$



Observe that the function of a single variable

$$g(x) = f(x, b+\Delta y)$$

is defined on the interval $[a, a+\Delta x]$ and $g'(x) = \frac{\partial f}{\partial x}(x, b+\Delta y)$. If we apply the Mean Value Theorem to g, we get

$$g(a+\Delta x) - g(a) = g'(u)\Delta x$$

where u is some number between a and $a+\Delta x$. In terms of f, this equation becomes

$$f(a+\Delta x, b+\Delta y) - f(a, b+\Delta y) = \frac{\partial f}{\partial x}(u, b+\Delta y)\Delta x.$$

This gives us an expression for the first part of the right side of Equation (4.6). For the second part we let $h(y) = f(a,y)$. Then h is a function of a single variable defined on the interval $[b, b+\Delta y]$ and $h'(y) = \frac{\partial f}{\partial y}(a,y)$. A second application of the Mean Value Theorem then gives

$$h(b+\Delta y) - h(b) = h'(v)\Delta y,$$

where v is some number between b and $b+\Delta y$. In terms of f, this becomes

$$f(a, b+\Delta y) - f(a,b) = \frac{\partial f}{\partial y}(a,v)\Delta y.$$

Now we substitute these expressions into Equation (4.6) and obtain

$$\begin{aligned}
\Delta z &= \frac{\partial f}{\partial x}(u,b+\Delta y)\Delta x + \frac{\partial f}{\partial y}(a,v)\Delta y \\
&= \frac{\partial f}{\partial x}(a,b)\Delta x + \frac{\partial f}{\partial x}(u,b+\Delta y)\Delta x - \frac{\partial f}{\partial x}(a,b)\Delta x + \frac{\partial f}{\partial y}(a,b)\Delta y + \frac{\partial f}{\partial y}(a,v)\Delta y - \frac{\partial f}{\partial y}(a,b)\Delta y \\
&= \frac{\partial f}{\partial x}(a,b)\Delta x + \left[\frac{\partial f}{\partial x}(u,b+\Delta y) - \frac{\partial f}{\partial x}(a,b)\right]\Delta x + \frac{\partial f}{\partial y}(a,b)\Delta y + \left[\frac{\partial f}{\partial y}(a,v) - \frac{\partial f}{\partial y}(a,b)\right]\Delta y \\
&= \frac{\partial f}{\partial x}(a,b)\Delta x + \frac{\partial f}{\partial y}(a,b)\Delta y + \varepsilon_1\Delta x + \varepsilon_2\Delta y,
\end{aligned}$$

where

$$\begin{aligned}
\varepsilon_1 &:= \frac{\partial f}{\partial x}(u,b+\Delta y) - \frac{\partial f}{\partial x}(a,b), \\
\varepsilon_2 &:= \frac{\partial f}{\partial y}(a,v) - \frac{\partial f}{\partial y}(a,b).
\end{aligned}$$

Since $(u,b+\Delta y)\to(a,b)$ and $(a,v)\to(a,b)$ as $(\Delta x,\Delta y)\to(0,0)$ and since $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are continuous at (a,b), we see that $\varepsilon_1\to 0$ and $\varepsilon_2\to 0$ as $(\Delta x,\Delta y)\to(0,0)$. Therefore f is differentiable at (a,b). □

Example 4.7. In Example 3.1, we found the partial derivatives of the function $f(x,y)=x^3+x^2y-2y^2$ were

$$\begin{aligned}
\frac{\partial f}{\partial x} &= 3x^2+2xy^3 \\
\frac{\partial f}{\partial y} &= 3x^2y^2-4y.
\end{aligned}$$

Since these are continuous everywhere, by Theorem 4.5 $f(x,y)$ is differentiable for all (x,y).

5 Chain Rule

Recall that the Chain Rule for real-valued functions of a single variable gives the rule for differentiating a composite function: If $y = f(x)$ and $x = g(t)$, where f and g are differentiable functions, then y is indirectly a differentiable function of t and

$$\frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt}.$$

For functions of more than one variable, the Chain Rule has several versions, each of them giving a rule for differentiating a composite function. The version we will need in this course deals with the case where $z = f(x,y)$ and each of the variables x and y is, in turn, a function of a variable t. This means that z is indirectly a function of t, $z = f(g(t),h(t))$, and the Chain Rule gives a formula for differentiating z as a function of t. We assume that f is differentiable (Definition 4.3). Recall that this is the case when $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are continuous (Theorem 4.5).

Theorem 5.1 (Chain Rule)**.** *Suppose $z = f(x,y)$ is a differentiable function of x and y, where $x = g(t)$ and $y = h(t)$ are both differentiable functions of t. Then z is a differentiable function of t and*

$$\frac{dz}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}.$$

Proof. If we change t to $t+\Delta t$, then this produces a change

$$\Delta x = g(t+\Delta t) - g(t)$$
$$\Delta y = h(t+\Delta t) - h(t)$$

in x and y, and therefore a change

$$\Delta z = f(x+\Delta x, y+\Delta x) - f(x,y)$$

in z. Since f is differentiable at (x,y), from Definition 4.3,

$$\Delta z = \frac{\partial f}{\partial x}\Delta x + \frac{\partial f}{\partial y}\Delta y + \varepsilon_1\Delta x + \varepsilon_2\Delta y,$$

where $\lim_{(\Delta x,\Delta y)\to(0,0)} \varepsilon_1 = \lim_{(\Delta x,\Delta y)\to(0,0)} \varepsilon_2 = 0$. Dividing both sides of this equation by Δt, we have

$$\frac{\Delta z}{\Delta t} = \frac{\partial f}{\partial x}\frac{\Delta x}{\Delta t} + \frac{\partial f}{\partial y}\frac{\Delta y}{\Delta t} + \varepsilon_1\frac{\Delta x}{\Delta t} + \varepsilon_2\frac{\Delta y}{\Delta t}.$$

If we now let $\Delta t \to 0$, then $\Delta x = g(t+\Delta t) - g(t) \to 0$ because g is differentiable and therefore continuous. Similarly, $\Delta y \to 0$. This, in turn, means that $\varepsilon_1 \to 0$ and $\varepsilon_2 \to 0$, so we have

$$\begin{aligned}
\frac{dz}{dt} &= \lim_{\Delta t\to 0}\frac{\Delta z}{\Delta t} \\
&= \frac{\partial f}{\partial x}\lim_{\Delta t\to 0}\frac{\Delta x}{\Delta t} + \frac{\partial f}{\partial y}\lim_{\Delta t\to 0}\frac{\Delta y}{\Delta t} + \left(\lim_{\Delta t\to 0}\varepsilon_1\right)\lim_{\Delta t\to 0}\frac{\Delta x}{\Delta t} + \left(\lim_{\Delta t\to 0}\varepsilon_2\right)\lim_{\Delta t\to 0}\frac{\Delta y}{\Delta t} \\
&= \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} + 0\cdot\frac{dx}{dt} + 0\cdot\frac{dy}{dt} \\
&= \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}.
\end{aligned}$$

□

Remark 5.2. Since we often write $\frac{\partial z}{\partial x}$ in place of $\frac{\partial f}{\partial x}$ (and similarly for y), we can rewrite the Chain Rule in the form

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}.$$

Example 5.3. If $z = x^2y + 3xy^4$, where $x = \sin(2t)$ and $y = \cos(t)$, then

$$\begin{aligned}\frac{dz}{dt} &= \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt} \\ &= (2xy + 3y^4)2\cos(2t) + (x^2 + 12xy^3)(-\sin(t)) \\ &= (2\sin(2t)\cos(t) + 3\cos^4(t))2\cos(2t) + (\sin^2(2t) + 12\sin(2t)\cos^3(t))(-\sin(t)).\end{aligned}$$

Exercise 5.4. Use the Chain Rule to find $\frac{dz}{dt}$ if $z = \cos(x+4y)$, $x = 5t^4$, $y = \frac{1}{t}$.